

Replication, evaluation and domain transfer

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1. Why this architecture

MK-UNet is a useful test of whether a compact model can remain experimentally useful when both data and compute are limited. Its encoder combines multiple depthwise kernels with pointwise channel mixing. Channel and spatial attention refine decoder features, while grouped gates modulate skip connections. The final full-resolution output alone is optimized for binary segmentation [1,2].

The screening experiments keep those learned modules unchanged. ClinicDB tests whether the architecture can be trained under the paper's intended optimization recipe on consumer hardware. Massachusetts tests whether it remains useful when the foreground changes from a polyp to many small building footprints. Each domain has its own learned weights; the experiment transfers an architecture, not a semantic label mapping.

2. Reference experiment and replication finding

ClinicDB test measure	Observed
Paper five-run mean Dice [1]	93.48%
Single-run native Dice, min-max normalized	89.59%
Single-run native Dice, fixed sigmoid threshold	89.59%
Native mean IoU, fixed sigmoid threshold	83.62%
Checkpoint selected on validation	Epoch 58

The signed gap to the reference is -3.89 percentage points. Min-max normalization changes the current mean by less than 0.01 point, so it does not explain that gap. This attempt preserves the published optimizer, loss, channels, multiscale inputs and training duration, while changing precision and correcting evaluation details. A mismatch could reflect optimization variance, protocol drift, or evaluation differences. Without repeated runs and a harmonized evaluator, it cannot establish that the published result is wrong.

3. What changed outside medicine

Massachusetts uses all 137 training tiles, all four validation tiles and all ten held-out test tiles. Tiles are resized to 512 x 512 and divided into four non-overlapping 256 x 256 patches. The selected checkpoint achieves native mean Dice 71.95% and mean IoU 56.25%. The adaptation is supervised training from scratch, using the same network, binary structure loss, and optimizer. It uses one scale to make its compute budget explicit.

Building footprints distribute foreground across many small components. Here, held-out building masks average 18.61% foreground pixels, versus 6.85% for the polyp masks; sparsity alone cannot explain domain difficulty. Downsampling can erase narrow roof details before the model sees them; this is an input limitation, not merely a decoder limitation. Stitching four patches also exposes seam artifacts. A future resolution experiment should compare this inexpensive preprocessing with native-resolution crops on the same tile split.

Working code and recorded runs

Bottlenecks and reliability evidence

4. Execution bottlenecks and controls

Activation memory remained a constraint despite only 315,566 parameters. ClinicDB uses mixed precision and three scale passes; buildings use FP32 and one scale. Their elapsed times therefore do not measure comparable architectural efficiency.

An early diagnostic building run erased red labels through grayscale conversion, producing invalid empty-mask scores. It was stopped and restarted with red-channel decoding after every mask was checked for black/red encoding. Invalid logs are preserved but excluded from reported metrics. Filename pairing, shape checks, image hashes and native masks provide the remaining data controls.

5. Predeclared stress tests

After supervised checkpoint selection, both models were tested under fixed brightness reduction (0.65), gamma (1.8), Gaussian noise (standard deviation 0.08 on [0,1]) and Gaussian blur (sigma 1.5). These are synthetic appearance shifts, not a claim of inter-hospital or inter-sensor validation. An exploratory one-step update changes only BatchNorm affine parameters, keeps source running statistics frozen, and resets before each case. A gate accepts a candidate only if horizontal-flip Jensen-Shannon disagreement decreases by at least 1e-6 and its predicted foreground area changes by no more than five percentage points.

Domain / condition	Frozen Dice	Gated Dice	Accepted
ClinicDB / clean	89.59%	89.57%	19/62
ClinicDB / brightness_0.65	80.30%	80.32%	27/62
ClinicDB / gamma_1.8	63.78%	63.81%	17/62
ClinicDB / noise_0.08	24.11%	24.10%	26/62
ClinicDB / blur_1.5	88.89%	88.89%	21/62
Buildings / clean	71.95%	71.95%	0/10
Buildings / brightness_0.65	59.13%	59.13%	0/10
Buildings / gamma_1.8	52.50%	52.50%	0/10
Buildings / noise_0.08	5.34%	5.34%	0/10
Buildings / blur_1.5	9.64%	9.65%	1/10

Across 310 polyp decisions, ungated candidates caused eight losses >1 Dice point; the gate retained one and accepted 110 candidates. For buildings, only 1/50 was accepted: blur benefit fell from +0.526 to +0.003 points. The gate sacrificed benefit while permitting harm. This is a specified BN-affine pilot, not full Tent/EATA/SAR. Per-case records include bootstrap intervals and measured overhead.

6. Interpretation and next experiment

Next, compare native-resolution crops with this low-resolution baseline, repeat seeds, and measure harmful acceptance and useful rejection at matched computation. These runs establish execution feasibility and expose reliability failures; they do not establish clinical usefulness or superiority over current adaptation methods.

References: [1-12] are listed in the accompanying proposal. Working code and recorded runs

Learning behavior and failure cases

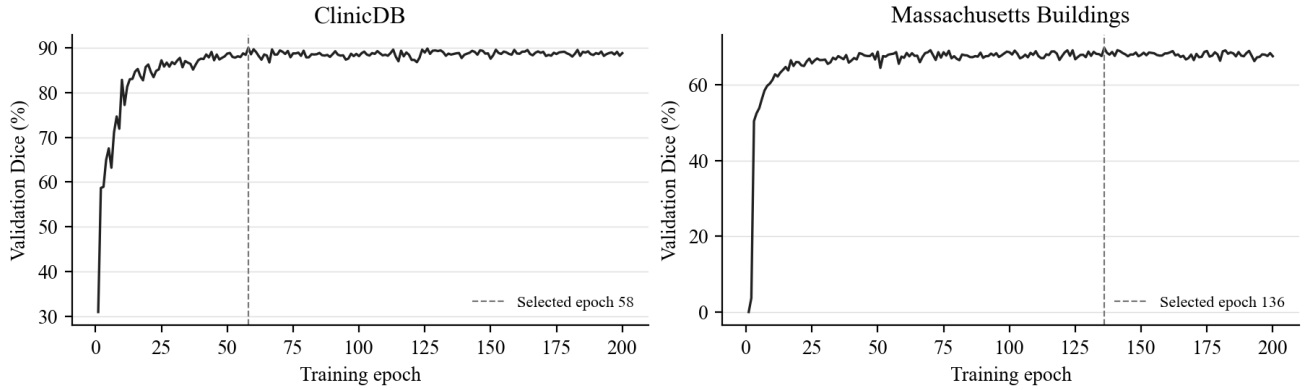


Figure 1. Validation trajectories over the complete 200-epoch runs. Dashed lines mark the selected checkpoints. The source and target selection metrics have different normalization conventions and should not be compared as equivalent scores.

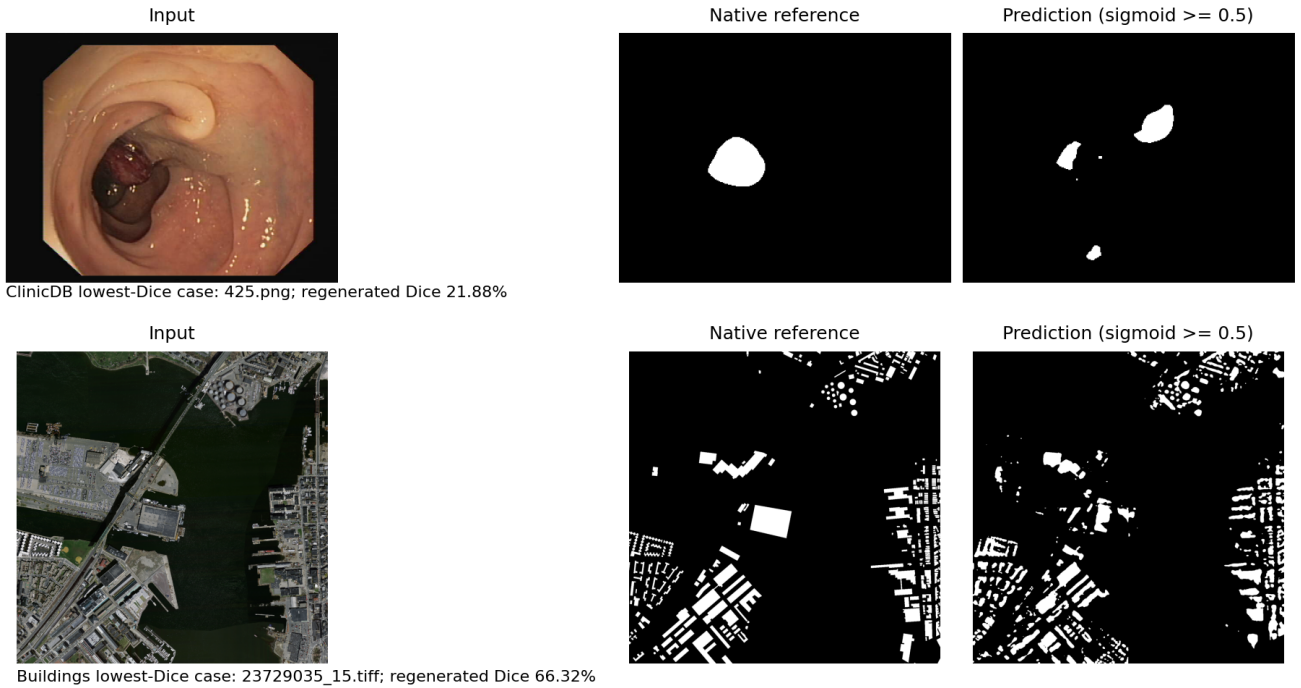


Figure 2. Lowest native fixed-threshold Dice test case in each domain, selected explicitly from the saved per-case results. Predictions were regenerated from best.pt and their Dice recomputed; small inference-rounding differences are recorded in the code archive. Display images are scaled to fit the page; the metric uses the original mask resolution.

These examples are evidence of failure modes in the tested pipeline, not a diagnosis of their cause. The proposed resolution and boundary ablations would test whether lost input detail, patch seams or adaptation decisions explain them.